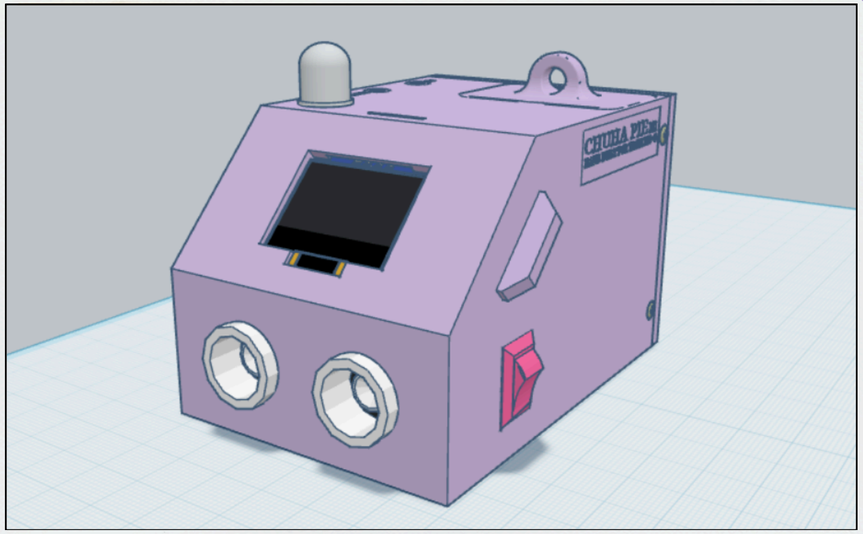
PRODUCT MANUAL 22/03/2026

GITHUB

CHUHA PIE v1

HAND BUILT 1/1 FOR YASHI PIE <3



Oink Oink!

LEGENDS:
Single Tap →.
Double Tap →..
Triple Tap →...
Long Touch → —

Home Screen

— → **Pet** the bot <3

... → Open ***Settings***
- . → Cycle b/w *Internal options*
- .. → Cycle b/w *various Settings*

→ ***Home Screen***

.. → **Bonk ;)**

. → Enter ***Feature Cycle*** Mode

Date & Time — → ***Stopwatch*** — → ***Pomodoro*** — → ***Spotify***
- Date & Time: . → Cycle b/w 12/24 Hr Format
- Stopwatch: . → Pause/Resume; .. → Reset
- Pomodoro: . → Cycle b/w options → *(Auto starts in 10 sec.)* → *Play/Resume*; .. → Reset (Pomodoro Menu)
- Spotify: . → Play/Pause; .. → Next Song; ... → Previous Song

— → ***Length Measurement Tool*** — → ***Temperature Display*** — → ***Humidity Display*** — → ***Home Screen***

Long pressing (>4s) in any Mode leads directly to the Home Screen

⚠ CAUTION
The Switch can be used to trigger *Deep-Sleep mode* on ESP32 which turns-off the device even with USB cable plugged in.

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GETTING STARTED

What you'll need:

1. A computer with USB port.
2. Arduino IDE.
3. USB cable (Micro-USB) - Supplied with the device.
4. WiFi connection to use Spotify Controls and sync time.
5. Spotify Premium account (for Spotify feature).

STEP-1: GET THE CODE

1. Go to the GitHub link - ***github.com/exorev07/Chuha-Pie-Desk-Companion.git***
2. Download ZIP.
3. Extract the ZIP to access the code.
   OR if you have Git installed: git clone.

STEP-2: SET UP ARDUINO IDE

1. Open Arduino IDE
2. Go to File → Preferences
3. Under Additional Boards Manager URLs, paste:

   https://raw.githubusercontent.com/espressif/arduino-%20esp32/gh-pages/package_esp32_index.json

4. Go to Tools → Board → Boards Manager, search esp32, install it
5. Go to Tools → Manage Libraries, search and install:
   - Adafruit SSD1306
   - Adafruit GFX Library
   - DHT sensor library by Adafruit
   - FluxGarage RoboEyes

GITHUB

STEP-3: CONFIGURE YOUR CREDENTIALS

1. Open the extracted folder & open **Main/secrets.h** in Arduino IDE
2. Find these lines & fill them in:

```
#define WIFI_SSID_VAL  "Enter 2.4Ghz SSID here"
#define WIFI_PASSWORD_VAL "ENTER WIFI Password"
```

STEP-4: SET UP SPOTIFY

1. Go to developer.spotify.com/dashboard and log in
2. Click Create App - name it anything, set redirect URI to *http://localhost:8888/callback*
3. Copy your ***Client ID*** and ***Client Secret, Value*** into **secrets.h**:

```
#define SPOTIFY_CLIENT_ID_VAL "Paste here"
#define  SPOTIFY_CLIENT_SECRET_VAL "Paste here"
```

4. To get your Refresh Token:
   Run ***spotify_get_refresh_token.py*** from the Product Repo.
5. Log in with your Spotify account and grant permissions
6. Copy the Refresh Token and paste it into **secrets.h**:

```
#define SPOTIFY_REFRESH_TOKEN_VAL "Paste here"
```

STEP-5: UPLOAD TO THE DEVICE

1. Plug in Chuha Pie via USB
2. In Arduino IDE go to Tools → Board → select ESP32 Dev Module or DOIT ESP32 Devkit V1
3. Go to Tools → Port → select the COM port that appears (e.g. COM3 on Windows)
4. Click the Upload button.
5. Wait for IDE to upload and when it displays "connecting..." press and hold boot button on ESP32 and the Device will restart automatically.

Auto Brightness

The device adjusts screen brightness automatically based on time of day
7:00 AM - 7:00 PM → *Full brightness (100%)*
7:00 PM - 7:00 AM → *Dimmed (50%)*
If the device can't connect to NTP server, it defaults to *75% brightness*.